

Sales Analytics Dashboard Using Python and Power BI

1. Project Overview

The **Sales Analytics Dashboard Using Python and Power BI** is a data analytics project developed to analyze sales data and understand business performance.

The project uses **Python** for data cleaning, data analysis and visualization, while **Microsoft Power BI** is used to create an interactive dashboard.

The dashboard provides useful information about total sales, total profit, total orders and total quantity. It also helps to analyze sales based on region, category, product and month.

The main purpose of this project is to convert raw sales data into meaningful information that can help businesses understand their performance and make better decisions.

2. Problem Statement

Businesses generate large amounts of sales data every day. Analyzing this data manually can be difficult and time-consuming.

The main problem is to analyze sales data efficiently and identify important information such as:

- Which region has the highest sales?
- Which category generates the highest profit?
- Which products are sold in higher quantities?
- How do sales change month by month?
- Which areas require improvement?

Therefore, this project provides a simple and interactive **Sales Analytics Dashboard** to solve these problems.

3. Project Objectives

The main objectives of this project are:

1. To clean and preprocess the sales dataset.
2. To analyze sales and profit using Python.

3. To identify important sales trends and patterns.
4. To create meaningful charts and visualizations.
5. To develop an interactive Power BI dashboard.
6. To provide useful business insights and recommendations.

4. Dataset Description

The dataset contains sales-related information such as order details, customer details, products, categories, regions, quantity, sales and profit.

Main Columns

- **Order_ID** – Unique order identification number.
- **Order_Date** – Date on which the order was placed.
- **Customer_ID** – Unique customer identification number.
- **Product** – Name of the product.
- **Category** – Product category.
- **Region** – Sales region.
- **State** – State where the order was placed.
- **Quantity** – Number of products sold.
- **Sales** – Sales amount.
- **Profit** – Profit generated from the sale.
- **Payment_Mode** – Payment method used by the customer.
- **Customer_Type** – Type of customer.

5. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Jupyter Notebook

- Microsoft Power BI
- Git
- GitHub

6. Data Cleaning Process

The following steps were performed during data cleaning:

1. Loaded the dataset into Python.
2. Checked the number of rows and columns.
3. Checked data types of all columns.
4. Identified missing values.
5. Handled missing categorical values.
6. Checked for duplicate records.
7. Removed duplicate records where required.
8. Converted the Order_Date column into proper date format.
9. Prepared the cleaned dataset for analysis and visualization.

7. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the sales dataset and identify important patterns.

The analysis focused on:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Region-wise Sales
- Category-wise Sales
- Category-wise Profit
- Monthly Sales

- Product-wise Quantity

8. Data Visualizations

Five main visualizations were created for the project:

1. **Sales by Region** – Bar chart showing sales generated by different regions.
2. **Monthly Sales Trend** – Line chart showing changes in sales over different months.
3. **Profit by Category** – Bar chart showing profit generated by each category.
4. **Sales by Category** – Bar chart showing sales for different product categories.
5. **Quantity Sold by Product** – Bar chart showing the quantity sold for different products.

9. Power BI Dashboard

An interactive **Sales Analytics Dashboard** was created using Microsoft Power BI.

KPI Cards

The dashboard contains four KPI cards:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity

Dashboard Visualizations

The following charts are included:

- Sales by Region
- Monthly Sales Trend
- Sales by Category
- Profit by Category

- Quantity by Product

Dashboard Filters

Five slicers were added to make the dashboard interactive:

- Year
- Region
- Category
- Payment Mode
- Customer Type

These filters allow users to view specific parts of the sales data easily.

10. Key Business Insights

The dashboard helps identify important business insights such as:

1. Sales performance can be compared across different regions.
2. Monthly sales trends help identify periods of high and low sales.
3. Category-wise analysis helps identify profitable product categories.
4. Product-wise quantity analysis helps identify popular products.
5. Profit analysis helps businesses understand which categories contribute more to profitability.
6. Interactive filters allow users to analyze sales from different perspectives.

11. Business Recommendations

Based on the analysis, the following recommendations can be made:

1. Focus more on high-performing regions.
2. Promote products with higher sales and quantity.
3. Improve the performance of low-selling categories.
4. Analyze monthly sales trends to plan future sales strategies.
5. Use Power BI dashboards regularly for better business decision-making.

12. Conclusion

The **Sales Analytics Dashboard Using Python and Power BI** successfully demonstrates how raw sales data can be transformed into useful business information.

Python was used for data cleaning, analysis and visualization, while Power BI was used to create an interactive dashboard with KPIs, charts and filters.

The project helps in understanding sales, profit, quantity and regional performance. It also demonstrates the practical use of **Data Analytics and Business Intelligence tools** for solving real-world business problems.

13. Project Structure

Sales-Analytics-Project/

- |— README.md
- |— Sales_Analytics.ipynb
- |— Sales_Analytics_Week6_Dataset.csv
- |— Visualizations/
 - | |— sales_by_region.png
 - | |— monthly_sales.png
 - | |— profit_by_category.png
 - | |— sales_by_category.png
 - | |— quantity_by_product.png
- |— PowerBI/
 - | |— Sales_Analytics_Dashboard.pbix
- |— Report/
 - | |— Final_Project_Report.pdf

14. Author

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